5. Probability

1. A card is drawn from a well-shuffled pack of 52 cards. Find the probability that it is (i) a king (ii) a red card (iii) not a face card.

2. Two unbiased coins are tossed once. Find the probability of getting (i) exactly one head (ii) at least one tail.

3. A die is thrown twice. Find the probability that the sum of the numbers is 7.

4. A bag contains 5 red balls, 3 blue balls and 2 green balls. One ball is drawn at random. Find the probability that it is (i) red (ii) not green.

5. From a group of 40 students, 28 like mathematics and 17 like science. If 12 students like both, find the probability that a randomly chosen student likes (i) only mathematics (ii) at least one of the two subjects.

6. The probabilities of three mutually exclusive events A, B and C are 0.3, 0.25 and 0.15 respectively. Find the probability that (i) none of them occurs (ii) at least one occurs.

7. The following data shows number of defective bulbs in 100 boxes: 0 defectives – 25 boxes, 1 defective – 30 boxes, 2 defectives – 20 boxes, 3 defectives – 15 boxes, 4 defectives – 10 boxes. If one box is chosen at random, find the probability that it has (i) exactly 2 defective bulbs (ii) at most 1 defective bulb.

8. A boy has 3 white shirts and 5 coloured shirts. He randomly picks one. Find the probability that it is coloured.

9. A game consists of spinning a wheel with numbers 1 to 10. If the wheel stops at a prime number, a player wins ₹50, otherwise loses ₹20. What is the expected gain or loss in one spin?

10. The probability that a student passes in Mathematics is 0.7 and in Science is 0.8. Assuming independence, find the probability that the student (i) passes in both subjects (ii) passes in at least one subject.